SafeInvest: Cost-Aware Robo-Advice with Explicit Risk Limits and Human Review

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Abstract

We build a transparent, cost-aware robo-advisory process for the SafeInvest synthetic scenario set. For suitability, we use real purchasing-power growth, ordinary return volatility, maximum real drawdown and a CVaR tail measure. A five-profile by three-product training screen assigns Profile A to human review, B to Conservative, C to Balanced, D to human review and E to Growth. For the Balanced customer we build V2b, an annual exact-fee policy that holds, restores, makes a bounded protective trade, or enters an absorbing human-review/hold state. Against the required one-third equal-weight baseline on the 60 final scenarios, V2b lowers median real CAGR from 3.38% to 2.35% and P10 growth from 1.47% to 0.87%, but lowers median real MDD from 22.29% to 20.02%, P90 MDD from 39.36% to 28.69% and breaches from 6/60 to 2/60; it trades in 21.22% of decision periods versus 100% for the baseline. A transaction-fee stress (0–1%) leaves the main recommendation unchanged on development data; a pooled-support change forces an initial Review, and +2 percentage points of inflation raises breaches to 21/120. Applied unchanged to the other two assigned profiles on development data, the same rule declines automatic funding at inception for B and E: the safeguard extends as a funding gate, and only C receives bounded autonomous service. The result is a risk–growth–service trade-off with an explicit human-review safeguard, not a demonstration of fully autonomous 30-year advice.

1 Data, accounting and evaluation split

The supplied file contains 180 independent synthetic scenarios, each with 30 annual periods of nominal equity, bond and bill returns and inflation. Returns and inflation are decimals; real wealth is nominal wealth divided by cumulative CPI. The data are synthetic and based on historical U.S. series; they are not historical observations or forecasts.

Every account starts with 100 units in bills and has no later deposits or withdrawals. Portfolios are long-only and fully invested. A trade charges 0.25% of each destination amount moved, counted once, deducted before that period's returns; the initial allocation is charged the same way. There are no taxes, other fees or terminal liquidation costs. A decision for period t uses only observations through $t-1$ of the same scenario.

Account model. Formally, an account is a vector $x_t \in \mathbb{R}^3_{\geq 0}$ of nominal asset values at the start of period t, with cumulative CPI c_t ($c_1 = 1$) and real wealth $v_t = \mathbf{1}^\top x_t / c_t$. A decision replaces x_t by post-trade holdings $\tilde{x}_t$ satisfying the self-financing identity

$$\mathbf{1}^\top \tilde{x}_t + \phi_t = \mathbf{1}^\top x_t, \qquad \phi_t = 0.0025\, m_t, \tag{1}$$

where m_t is the total destination amount moved, counted once. Returns then act: $x_{t+1} = \tilde{x}_t \odot (1 + r_t)$ and $c_{t+1} = c_t(1 + \pi_t)$. The running real peak and maximum drawdown are

$$P_t = \max_{s \leq t} v_s, \qquad D_t = \max_{s \leq t} (1 - v_s / P_s), \tag{2}$$

and a *breach* is the absorbing event $D_t > L$ for the customer's stated loss tolerance L. The decision at t is a function of the state (x_t, c_t, P_t, D_t) and of observations through $t-1$ only.

Original scenarios 1–120 are used for development and adjustment; scenarios 121–180 are the final evaluation set. The final 60 were exposed earlier in the project, so the final comparison is a fixed-policy, *nonblind* repetition, not a fresh blind sample. We made no parameter, target or admission change after the final run. This disclosure is material for interpreting every final number.

2 Question 1: suitability, products and profiles

2.1 Measures

Growth is the net real annualised log growth $G_H = [\ln(V_H/\mathrm{CPI}_H) - \ln(100)]/H$, with CAGR $= \exp(G_H) - 1$. Ordinary fluctuation is the sample standard deviation of annual net real returns. Large loss is the maximum real drawdown from the complete original peak, using annual endpoints; it is positive and increasing in loss. Tail severity is the empirical CVaR90 of the MDD distribution (and P90 MDD). All measures use purchasing power, not nominal wealth. Annual endpoints do not observe intrayear highs or lows. The questionnaire loss tolerance L is a response and a review trigger, never a guaranteed portfolio limit.

2.2 Products

Table 1: The three named products. Bill floors are target conventions for the liquidity need, not live cash reserves.

Product	Equity	Bond	Bill
Conservative	25%	30%	45%
Balanced	45%	5%	50%
Growth	65%	5%	30%

The target bill floor by liquidity is 30% for High, 10% for Medium and 0% for Low. A target floor is not a withdrawal guarantee or a persistent live-cash balance.

2.3 Onboarding rule

Figure 1 gives the one-page decision flow. In words: (1) take H, L and liquidity; a missing or materially changed cash need is a review trigger; (2) fix the three products and bill floors; (3) for each profile's own horizon, evaluate each product on 1–120 with exact fees and annual target restoration; retain a product only if its empirical MDD $> L$ frequency is at most 10% and its target bill meets the liquidity floor; pool the profile evidence into a common menu; (4) if no product is feasible, output Review; (5) among feasible products retain training growth within 10^{-12} of the global maximum, then minimise expected loss excess, then CVaR90, then the declared candidate index; (6) assign the product. The 10% is a declared service reference, not a contest mandate or a calibrated probability.

2.4 The 5×3 development screen

Table 2 is the 15-cell development screen. Each cell reports median CAGR / P10 CAGR; median MDD / P90 MDD; and breaches out of 120. The bold cell is the frozen assignment. All selected B/C/E products pass the declared 10% training screen (B 9/120, C 9/120, E 6/120). A and D have no risk-feasible product: A's alternatives breach in 34–66 and D's in 67–104 of 120 paths, so the liquidity floor alone cannot justify an allocation.

Q1 onboarding decision flow

One-page suitability rule for Profiles A–E.

1. Inputs and review gate H, loss tolerance L, liquidity need. Missing or materially changed cash need → human review.

↓

2. Product menu and bill floors Conservative 25/30/45; Balanced 45/5/50; Growth 65/5/30. Bill floors: high .30, medium .10, low 0.

↓

3. Own-horizon training screen (1–120) Simulate each product with exact 0.25% fees and annual restoration. Require empirical MDD $> L$ frequency $\leq 10\%$ and target bill $\geq$ liquidity floor; retain the common menu. No feasible product → Review (A, D).

↓

4. Rank and assign Retain G within 10^{-12} of the global maximum; tie-break by expected excess, then CVaR90, then candidate index. Result: A Review, B Conservative, C Balanced, D Review, E Growth.

↓

5. Runtime safeguard Real annual MDD $> L$ or funded-support failure → absorbing human review/hold; no new automatic trades; existing holdings remain market-exposed.

Development decision trace

Profile	Risk-feasible	Joint	Menu	Assignment
A	0	0	0	**Review**
B	52	52	1	**Conservative**
C	93	93	2	**Balanced**
D	0	0	0	**Review**
E	157	92	3	**Growth**

What the numbers mean

A and D have no risk-feasible product; the liquidity floor alone is not enough. B/C/E have a common-menu candidate. The 10% screen is a declared service reference, not a contest mandate or a guarantee. On final 121–180, B/C static frequencies rise above 10% and E Growth has negative P10 growth; these adverse findings are disclosed, not repaired by changing assignments. Q1 static products are not the Q2 guarded service.

Source: frozen V2b registry; full-development 1–120 feasibility counts. Exact tie-break and hashes: `service_refinement_readiness_review.json`, `joint_liquidity_refit_protocol.json`.

Figure 1: One-page onboarding decision flow and development decision trace. The flow is a finite-menu rule; it is not a claim of global optimality. The 10% screen is an empirical training reference.

Table 2: Development 1–120 static product screen, 5 profiles × 3 products. Each cell: median CAGR / P10; median MDD / P90; breaches/120. Bold = frozen assignment. This is a product screen, not the Q2 guarded service.

Profile	Conservative 25/30/45	Balanced 45/5/50	Growth 65/5/30
A (H5, L10%, high)	3.17 / −1.59 5.56 / 14.90 34/120	4.36 / −1.85 8.59 / 20.53 49/120	5.10 / −2.73 11.94 / 27.46 66/120
B (H15, L25%, medium)	**3.04 / 0.10** **13.25 / 24.22** **9/120**	4.08 / 0.10 17.35 / 32.65 27/120	5.26 / 0.02 23.18 / 42.07 49/120
C (H30, L40%, low)	2.97 / 1.02 17.37 / 31.74 1/120	**4.03 / 1.44** **22.60 / 38.04** **9/120**	5.12 / 1.48 32.26 / 48.84 35/120
D (H25, L15%, medium)	3.19 / 0.90 16.71 / 29.37 67/120	4.31 / 1.34 20.90 / 35.91 92/120	5.33 / 1.41 29.18 / 47.90 104/120
E (H6, L35%, high)	2.99 / −1.74 8.08 / 18.35 0/120	3.75 / −2.33 10.29 / 23.11 2/120	**5.00 / −3.69** **13.89 / 31.18** **6/120**

2.5 Final assignments and own-horizon evidence

Table 3 evaluates the frozen assignments on final 121–180. A and D remain Review with all financial fields NULL; they are not replaced by bills, the baseline or a zero-risk portfolio. B/C/E use each profile's own horizon and loss tolerance.

Table 3: Final 121–180 own-horizon results for the frozen assignments. A/D are unfunded Review (NULL), not zero-risk investments.

Profile	Assignment	Target	Med. CAGR	P10	Med. MDD	P90	Breaches
A	Review	—	NULL	NULL	NULL	NULL	NULL
B	Conservative	25/30/45	3.21%	0.35%	14.09%	29.53%	11/60
C	Balanced	45/5/50	3.56%	1.48%	27.49%	43.54%	8/60
D	Review	—	NULL	NULL	NULL	NULL	NULL
E	Growth	65/5/30	4.51%	−1.58%	14.17%	31.34%	5/60

Briefly: A has a 5-year horizon, a 10% loss tolerance and high liquidity; no product passes the risk screen, so Review is the honest output. B is Conservative: the 15-year/25%/medium profile retains the lowest MDD and breach frequency among its three products, and its development Conservative row passes the 10% screen at 9/120. C is Balanced: it buys growth over Conservative, passes the development screen at 9/120, and the Q2 guard is the execution safeguard. D is Review: at 25 years, 15% loss tolerance and medium liquidity, every product breaches 67–104 of 120 development paths. E is the most debated assignment: Growth has the highest median CAGR (4.51%) and a 30% bill floor, but its final P10 is −1.58% and it breaches 5/60; the rule selects the highest-growth product that passes the declared 10% MDD screen, not the product with the best lower growth tail. This is a growth-for-tail trade-off, not a guarantee.

Adverse tails must remain visible. On final data, the static B/C frequencies are 18.3% and 13.3%, above the 10% training reference; E's P10 is negative. The 10% reference is a declared service convention, not a contest mandate; Q1 products are static annual references, not the Q2 guarded

service. C's V2b guard reduces final breaches to 2/60. We do not retroactively change the assignments to make the final table look better.

2.6 Does the safeguard extend to B and E?

The Q2 guard was developed for C. To test whether it is one coherent safeguard family or a C-only device, we applied the *unchanged* locked rule to B and E with each profile's own target, horizon and L on development scenarios 1–120 (a resubstitution diagnostic on data already used for development; no final scenarios and no new fitting).

Table 4: The unchanged execution gate at inception, development 1–120. The budget floor is the declared convention $(1 - L/2) \times v_1$ with $v_1 = 100$. Only C passes; B and E enter human review before any automatic investment.

Profile	Product	L	Initial floor	Gate decision
B	Conservative 25/30/45	25%	87.5	Pre-investment Review
C	Balanced 45/5/50	40%	80.0	Funded (60/60 on final data)
E	Growth 65/5/30	35%	82.5	Pre-investment Review

Under the calendar support (Section 3.1), no eligible initial allocation exists for the Conservative or Growth targets at their floors: every proposed allocation admits a modeled one-year outcome below the released headroom. The safeguard is therefore one family with three consistent layers: suitability review at onboarding (A, D), a funding gate before any automatic investment (B, E), and bounded autonomous service only where both gates pass (C). The static Q1 numbers of Table 3 are suitability references, never automatic execution approvals — and the gate confirms this by refusing unattended funding for B and E. Because no funded B/E guard paths exist, this is a statement about the gate's conservatism; we claim no executed breach reduction for these profiles. Protocol, independent audit (10,993 checks) and byte-exact isolated replay are in the supplementary package.

3 Question 2: the Balanced policy

3.1 Rule

For Profile C (Balanced, $H = 30$, $L = 40\%$, low liquidity) the strategic target is 45% equity, 5% bond and 50% bills. At each annual decision the policy, which we call V2b, can hold, restore the target, make a temporary protective exposure proposal, make one first mitigation, or enter human review. It does not change the strategic target: a protective proposal temporarily shifts actual holdings toward bills, and a review/hold freezes new automatic trades. A V3 target-update ablation was not selected because it had mixed growth/risk effects and higher fees; this is a development judgment, not proof that target updates never help.

Support and floor. Let $S_t = \{(1+r_t^{(j)})/(1+\pi_t^{(j)})\}_{j=1}^{120}$ be the *calendar support*: the real gross return vectors of the 120 development scenarios at the same calendar year t. For post-trade holdings y, the worst supported next-year real wealth is $\underline{W}_t(y) = \min_{g \in S_t} g^\top y / c_t$. The client floor is $F_t = (1 - L)P_t$; the budget floor releases only the fraction $\rho = 0.5$ of the remaining headroom,

$$B_t = \max\{F_t,\ v_t - \rho\,(v_t - F_t)\}. \tag{3}$$

Candidate menu and selection. Thirteen actions compete at each review: hold, and for each of four tilt levels $\tau \in \{1, 0.75, 0.5, 0.25\}$ toward the bill corner e_{bill}, the exact fee-inclusive segment from x_t toward $\tau w + (1 - \tau)e_{\text{bill}}$, sampled at its two support-feasible endpoints and midpoint. An action

a is eligible only if $\underline{W}_t(\tilde{x}_t(a)) \geq B_t$. Among eligible actions the policy maximizes the fee-net paired predicted log advantage over holding,

$$s(a) = \frac{1}{120}\sum_{j=1}^{120} \ln \frac{g_j^\top \tilde{x}_t(a)}{g_j^\top x_t}, \qquad g_j \in S_t, \tag{4}$$

and trades only when $s(a) > 0$ strictly; ties break toward holding and lower fees. The forecast is uniform over the same training year-t real gross returns, so the benefit test is exactly the estimated-benefit-justifies-cost condition required by the problem; nothing is fitted on the evaluation scenarios.

Protection, mitigation and review. If holding itself violates the floor, the same menu applies but predicted growth may be sacrificed: a protective trade is admitted only when it strictly improves the modeled worst supported outcome after exact costs, and is labelled risk reduction with a growth cost. If no eligible proposal exists, a one-time nine-action mitigation menu (hold; the four tilt endpoints at fractions $q \in \{0.5, 1\}$) chooses the action maximizing $\underline{W}_t$ net of exact fees, again only under strict improvement over hold, and the account then enters absorbing human review. A realized breach $D_t > L$ at any annual endpoint also triggers the absorbing pending-hold state. At inception the full target must pass the same support test, otherwise the account is never funded (the gate of Section 2.6). While pending, the app holds the existing holdings, continues market exposure and makes no new automatic trades.

3.2 Baseline and metrics

The required baseline is one-third equity, one-third bond and one-third bills, restored at every review, with the same 0.25% fee. We compare median CAGR, median MDD, P10 CAGR, P90 MDD, breaches, trade fraction and mean nominal fees. P10/P90 use the empirical inverse CDF; medians use the ordinary median. MDD uses annual endpoints from the original peak and does not measure intrayear losses.

Table 5: Final 121–180 comparison. The equal-weight baseline is the required one-third annual-restoration rule. "Same-target pending" restores the Balanced target annually until the first breach and then holds; "same-target annual" restores every year. Trade fraction counts periods with actual destination movement; mean fees are nominal units per initial 100.

Strategy	Med. CAGR	Med. MDD	P10	P90	Breaches	Trade	Fees
V2b	2.35%	20.02%	0.87%	28.69%	2/60	21.22%	0.353
Same-target pending	3.56%	27.49%	1.50%	43.54%	8/60	94.56%	1.018
Same-target annual	3.56%	27.49%	1.48%	43.54%	8/60	100%	1.067
Equal thirds	3.38%	22.29%	1.47%	39.36%	6/60	100%	0.979

V2b has the lowest tail risk and fees, but it also has lower median growth and lower P10 growth than the equal-weight baseline. It is not an all-metric dominance claim. Against equal thirds, the median MDD falls by 2.27 percentage points, P90 MDD by 10.68 points, breaches from 6 to 2, fees to about 36% and trading from 100% to 21.22%; median CAGR falls by 1.03 points and P10 by 0.60 points. The policy therefore buys drawdown and fee control with growth.

The mechanism matches the design. V2b spends most periods in hold or pending-hold, so compounding is slower in good states; but the floor test (3) forces de-risking exactly when the modeled one-year worst case threatens the released headroom, and the absorbing review stops the policy from adding new automatic exposure afterwards. The improvements therefore concentrate in the tail columns (P90 MDD, breaches, fees) while the growth columns deteriorate. Pending does not stop losses — holdings remain exposed, as the severe case below shows.

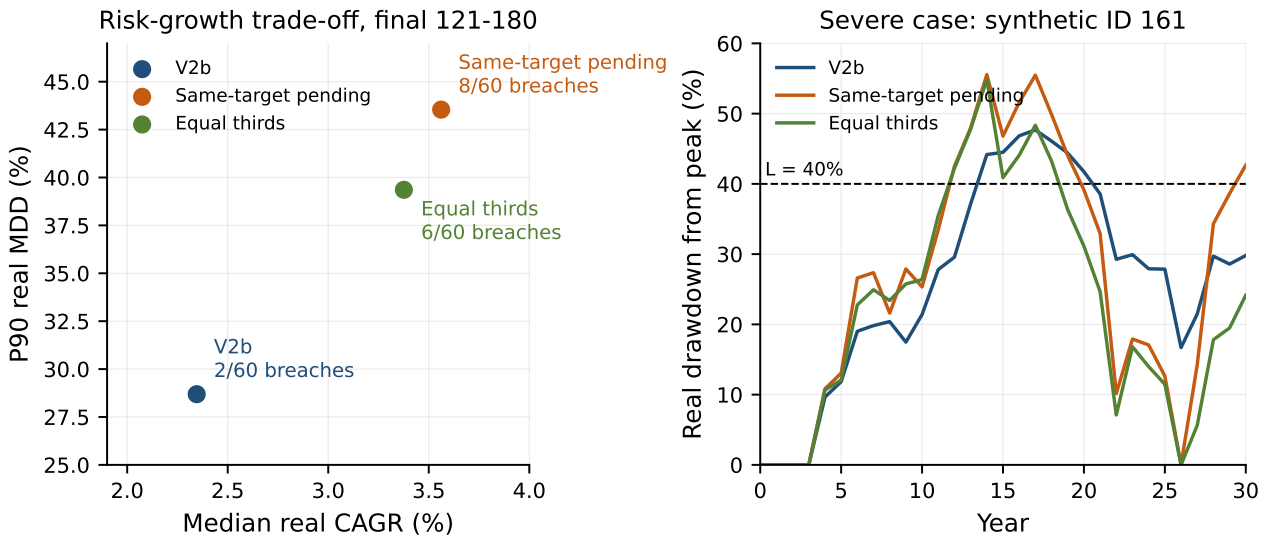


Figure 2: Left: final risk–growth trade-off. Right: real drawdown path of the preselected severe synthetic case ID 161. V2b reduces the drawdown relative to the same-target pending reference and equal thirds but still crosses the 40% limit.

3.3 Trade frequency

There are 382 trade periods out of 1,800 decision periods: $382/1800 = 21.22\%$, including the mandatory initial allocation. Excluding the initial allocation, $322/1740 = 18.51\%$. Ordinary autonomous service is a different metric: $350/1{,}800 = 19.44\%$ of account-years. All 60 paths enter human review by decision year 7 (54 in year 7); the app is a bounded assisted service, not a 30-year autonomous trader.

3.4 Severe case and customer explanation

The preselected severe case is synthetic ID 161, chosen by maximum main full-horizon MDD with smallest-ID tie-break. V2b has MDD 47.68%, CAGR −0.17%, first crossing at year 14, 20% trade fraction and 0.236 nominal fees. Same-target pending has 55.56%, −0.26%, year 12 and 40% trade; equal thirds has 54.84%, +0.32%, year 12 and 100% trade. V2b lowers drawdown versus both references but still breaches, and its growth in this path is worse than equal thirds. The two final breaches are IDs 132 and 161, at endpoints 23 and 14; both occur after the first referral. A one-year supported action is not a recursive or real-market safety guarantee.

The customer-facing explanation is 41 words:

> Your portfolio's risk buffer no longer supports our usual trading options. We make a protective trade only if the modeled worst outcome improves after costs, then pause automated trading for human review. Existing holdings remain invested and may continue losing value.

It identifies the protective trade, the review request and the continued market exposure. It does not promise recovery or an adviser response.

3.5 Do automated weight updates add value?

They add *risk-control* value, not growth value. The evidence supports temporary protective/review updates but not persistent strategic target updates. The final policy is deliberately conservative: 19.44% ordinary service, 2/60 breaches and all 60 paths referred by year 7. We do not describe this as a successful 30-year autonomous robo-advisor.

4 Question 3: reliability and safeguards

4.1 Transaction-fee stress

We test known fees 0%, 0.125%, 0.25%, 0.5% and 1% on the frozen V2b arm over development 1–120. All actors know the same fee; there is no refit or new selection. The main recommendation does not change: breaches stay 0/120 and median CAGR moves from 2.865% to 2.802%; P90 MDD stays near 28.98%. Trading and fees move modestly. This is a cost-magnitude test, not a test of fee misestimation, refitting or real-market safety.

Table 6: Development 1–120 stress tests. Pooled support and +2pp inflation are assumption changes; the final 121–180 sample is not used here.

Setting	V2b result	Main recommendation
Fee 0–1%	0/120 breaches; median CAGR 2.80–2.86%; P90 MDD ≈28.98%	unchanged
Pooled support	0 funded; initial Review; financial NULL	changes to Review
Inflation +2pp	21/120 breaches; median CAGR 0.75%; P10 −1.16%; P90 MDD 43.54%	suspend/review

4.2 Important limitation and trigger

The one-year support/forecast condition is empirical and finite. It is not a conditional return distribution, a calibrated loss probability, an intrayear stop-loss or a future guarantee. The final 121–180 sample was exposed earlier; it is not fresh blind validation. This is the main limitation.

The primary review trigger is a funded support-family failure or no feasible supported action. A protective trade is allowed only if the modeled worst next-real-wealth outcome improves after exact costs; then the account enters absorbing human review. An annual real MDD above the customer's L also triggers review. Model/input-integrity failure or a material questionnaire/cash-needs change triggers pre-investment review. While pending, the app holds existing holdings, makes no new automatic trades, continues actual market exposure, CPI, original peak/MDD and financial history, and notifies the customer and adviser. It does not assume an adviser response, a recovery reset or a liquidation.

5 Conclusion

The three products are Conservative 25/30/45, Balanced 45/5/50 and Growth 65/5/30. The onboarding rule assigns A Review, B Conservative, C Balanced, D Review and E Growth. For the Balanced customer, the V2b policy uses protective/review updates but does not enable persistent strategic target updates. The main safeguard is layered and consistent: a suitability screen at onboarding, the same unchanged support gate before any automatic funding (which admits only C), and an absorbing human review/hold after funded-support failure or MDD above L — with no new automatic trades while pending and existing holdings remaining exposed.

The final evidence is a trade-off, not a universal winner. V2b lowers P90 MDD, breaches and fees relative to equal weights, but also lowers median and P10 growth, and ordinary autonomous service is only 19.44% of account-years. The final 121–180 sample was previously exposed; no claim of fresh blind validation or 30-year autonomy is made. Within those limits, the result is a coherent, explainable and risk-aware assisted service.

Reproducibility. The fixed-table adapter reproduces all 14 final objects, 840 path rows and 505 saved array fields from the supplied data. The B/E funding-gate addendum has its own frozen protocol, an independent audit recomputing every figure directly from the saved physical accounts (10,993

checks) and an isolated replay matching all deterministic artifacts byte for byte (210 array fields). The repository's full test suite has 1,545 passing tests. These checks establish implementation and reproduction integrity; they do not establish predictive validity or a fresh blind test.

Use of AI tools

AI assistants including OpenAI Codex and DeepSeek were used for problem interpretation, method discussion, code drafting and debugging, synthetic testing, numerical review and English editing. Automated numerical audits and independent-agent reviews were used as checks, not as external human peer review. All AI-generated content, code, calculations and references were reviewed by the participant; Chengxuan YUAN, Team Lebmar (87), remains responsible for the submitted work and its disclosures.

Data attribution

The SafeInvest Synthetic Dataset is based on selected United States series from the Jordà–Schularick–Taylor Macrohistory Database, Release 6. It does not reproduce historical observations and is not a forecast. The dataset and problem statement are distributed for noncommercial use under CC BY-NC-SA 4.0; attribution and share-alike conditions are retained. No new blanket open-source licence is granted to the participant's original code.

References

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